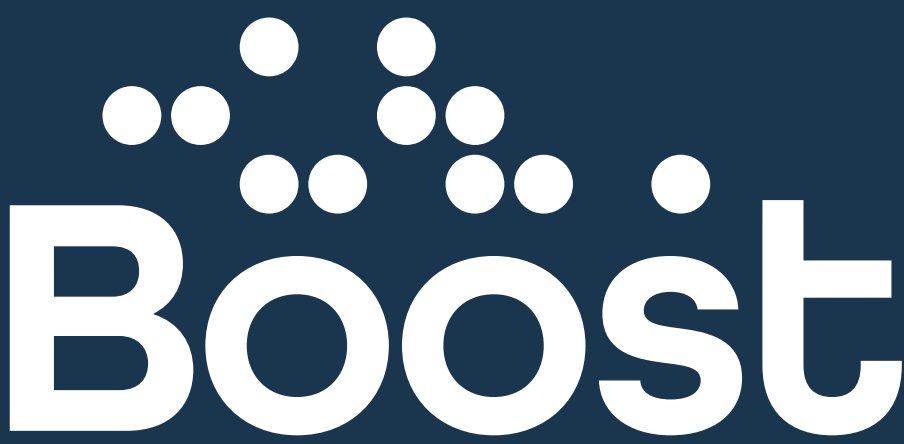
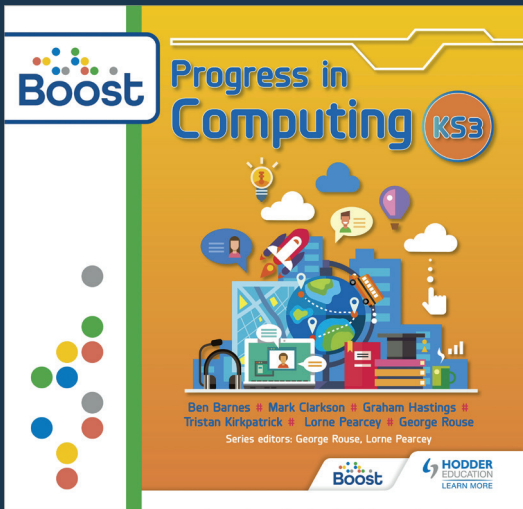
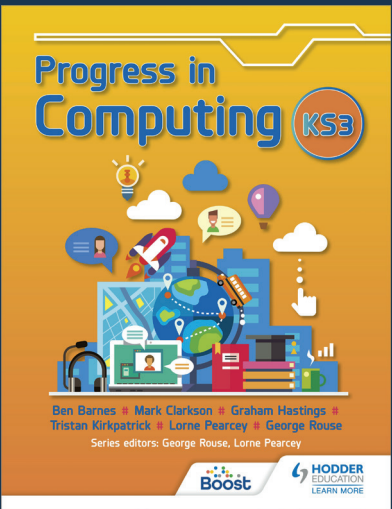
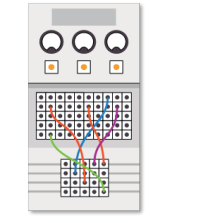
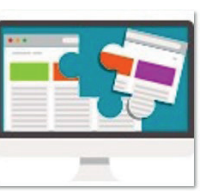
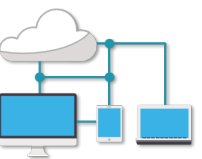


PROGRESS IN COMPUTING

Three-year progression pathway

This poster maps each *Progress in Computing* module to 5 core learning objectives. These objectives are designed to cover the entirety of the KS3 National Curriculum, in addition to developing digital literacy skills and preparing them for the range of KS4 computing options.



BY THE AGE OF 11, STUDENTS SHOULD BE ABLE TO: <i>Key Stage 2 National Curriculum for Computing</i>	• Use sequence, selection and repetition in programs • Select, use and combine a variety of software on a range of digital devices • Understand computer networks and how they provide multiple services • Use search technologies		• Detect and correct errors in algorithms and programs • Design, write and debug programs that accomplish specific goals		• Use technology safely, respectfully and responsibly
By the end of <i>Progress in Computing</i>, students should be able to:	<i>Know and understand the key concepts and principles of Computing</i>	<i>Apply knowledge and understanding of the key concepts and principles of Computing</i>	<i>Analyse problems in computational terms</i>	<i>Plan creative solutions to problems</i>	<i>Develop confident and responsible use of modern information technologies</i>
 Getting started	• Know the processes for logging into the school's network • Know the processes for sending and receiving emails • Understand how to save, rename and organise files • Understand how to access files stored in the cloud • Understand key principles of internet safety • Understand the qualities of vector and bitmap graphics	• Log into the school's network proficiently • Send and receive emails successfully, using appropriate language and content • Organise files and folders to facilitate ease of access and use • Demonstrate safe practices when using the Internet • Be able to create and manipulate images	• Identify the most appropriate tools to use when editing an image		• Demonstrate proficiency in using the school's network and computing facilities • Use image editing software with confidence
 Introducing spreadsheets	• Understand how to write basic formulae in a spreadsheet • Understand the concept of replication and the uses of relative and absolute cell referencing • Understand how to name cells and ranges within a spreadsheet • Understand how to write a range of basic functions including SUM, AVERAGE, MAX, MIN, COUNT and IF • Understand how to use conditional formatting • Understand how to use data in a spreadsheet to create graphs and charts	• Use a range of basic formulae to manipulate data • Use conditional formatting • Create graphs and charts to represent different types of information	• Identify the most appropriate functions to use when developing a spreadsheet for a particular purpose • Identify the most appropriate chart or graph to display different types of information		• Become proficient in the use of spreadsheets to handle data in a variety of situations • Interpret data from spreadsheets
 Computing: past, present and future	• Know about important figures in the development of computing • Understand Moore's Law and how computer technology has developed and changed over time • Know how to format documents • Understand the importance of aesthetics when presenting information and have an awareness of factors that can inhibit this	• Present knowledge about computing using word processing and presentation software • Use formatting appropriately • Ensure that work has been proofread and that spelling and grammar has been checked	• Select appropriate text and images for use in presentations	• Design presentations to convey information effectively	• Use word processing and presentation software to present information effectively
 Programming in Scratch	• Understand the concepts of sequencing, selection and iteration	• Develop working programs in Scratch	• Analyse the requirements of a program • Identify the processes needed to solve a problem	• Design programs in Scratch to solve specific problems	• Use Scratch confidently to solve a range of problems
 Computing components	• Know about and understand the function of a range of input and output devices • Know about and understand different types of memory and storage and their use		• Identify the correct input and output devices to use in a range of different situations		
 Programming in Python: sequence	• Understand a range of basic programming constructs in Python • Know how to print to the screen, perform calculations, take inputs and store them in suitably named variables	• Develop working programs in Python to solve specific problems	• Analyse the requirements of a program • Identify the processes needed to solve a problem	• Design programs in Python to solve specific problems	• Use Python confidently to write simple programs
 Advanced spreadsheets	• Understand the structure and use of a range of more advanced functions • Understand how to use validation to create dropdown lists • Know how to sort data and run simple queries • Understand the use of macros to automate processes and know how to record, edit and assign macros	• Use a range of more advanced functions within spreadsheets • Use validation within spreadsheets to minimise user error • Develop and use macros to automate aspects of spreadsheets	• Identify the most appropriate functions to use when developing a spreadsheet for a particular purpose	• Design spreadsheets for a range of purposes making use of a range of more advanced functions	• Use spreadsheets to handle data in a variety of situations proficiently • Interpret data from spreadsheets
 Algorithms	• Understand the concepts of abstraction, decomposition, pattern recognition and algorithms • Know how to read and develop flow diagrams	• Use the principles of abstraction and decomposition to produce algorithms to solve a range of problems • Write flow diagrams to sequence the steps involved in completing a task	• Analyse different approaches to solving problems	• Design algorithms to solve a range of computational problems	
 Programming in Python: selection	• Understand how to use selection in Python • Understand how to program condition-controlled loops in Python	• Develop working programs in Python to solve a range of problems	• Analyse the requirements of a program • Identify the processes needed to solve a problem	• Design programs in Python to solve a range of problems	• Use Python confidently to write a wider range of programs
 Internet safety, cyber security and encryption	• Understand a range of malware and the effects they have • Know what precautions to take to maintain safety online • Understand the role of encryption in maintaining safety online • Know about a range of ciphers	• Demonstrate safe practices when using the Internet • Use a range of ciphers to encrypt and decrypt text			• Use computer systems safely and confidently
 Binary and computer logic	• Understand binary and why it is used in computing • Know how to convert between denary and binary • Understand how binary is used to encode text and images • Understand the concept of AND, OR and NOT gates and their use in computer programs	• Carry out binary/denary conversions • Encode and decode text and images in binary	• Identify the output from simple logic circuits	• Design an app to meet a brief	
 Sound and video editing	• Understand how to use audio editing software • Know about a range of effects that can be applied to sound files • Understand how to use video editing software • Know how to combine images and sound	• Record and edit audio files • Develop and edit videos • Combine video and audio		• Create a plan to develop a video	• Use audio and video editing software with confidence • Review and evaluate work involving audio and video files
 Designing websites	• Know a range of basic HTML tags to control the presentation, styling and layout of information on a webpage • Know how to include images and links on a webpage • Understand the role of CSS • Appreciate the factors that contribute to a well-presented web page	• Hand-code basic web pages including text, images and hyperlinks • Use CSS to control the styling of a basic web page	• Identify the basic elements that are combined to produce a web page	• Design simple web pages	• Be able to hand-code basic web pages, with an appreciation of layout and styling • Review and evaluate work involving websites
 Networking and the Internet	• Understand how data is sent across a network • Know the role of a range of basic hardware involved in networking, such as switches • Understand the role of IP addresses • Understand domain names and DNS • Know about a range of Internet services				
 Programming in Python: iteration	• Understand how to program count-controlled loops in Python • Understand the concept of nested loops	• Develop working programs in Python to solve a wide range of problems	• Analyse the requirements of a program • Identify the processes needed to solve a problem	• Design programs in Python to solve a wide range of problems	• Use Python confidently to write a wide range of programs
 The ethics of computing	• Understand the role of algorithms in decision making • Understand the importance of intellectual property and copyright	• Make informed judgements about whether activities are morally acceptable or not • Ensure that copyright has not been infringed when using resources found online • Store data safely with regard to current legislation	• Consider the ethical implications of using modern information technologies	• Research resources online, being mindful of copyright considerations and acknowledging sources	• Use modern information technologies responsibly
 Projects (end-of-KS3 assessment) <i>Two projects chosen from a choice of three</i>		• Develop a substantive product using a range of software and techniques studied at KS3	• Analyse the requirements of a substantive product	• Plan a solution to develop a substantive product, including sub-tasks and timelines	• Use a range of software to produce a substantive product, with regard to copyright issues • Review and evaluate project work
By the end of <i>Progress in Computing</i>, students should be able to:	<i>Know and understand the key concepts and principles of Computing</i>	<i>Apply knowledge and understanding of the key concepts and principles of Computing</i>	<i>Analyse problems in computational terms</i>	<i>Plan creative solutions to problems</i>	<i>Develop confident and responsible use of modern information technologies</i>
AT THE END OF KEY STAGE 3, STUDENTS SHOULD BE ABLE TO: <i>Key Stage 3 National Curriculum for Computing</i>	• Understand several key algorithms that reflect computational thinking • Understand simple Boolean logic and its use in programming and circuits • Understand how numbers can be represented in binary and be able to carry out simple operations on binary numbers • Understand the hardware and software components that make up computer systems • Understand how data of various types can be represented and manipulated digitally	• Use a variety of programming languages, including a textual language, to solve a variety of computational problems • Undertake creative projects that involve selecting, using and combining multiple applications	• Design, use and evaluate abstractions that model the state and behaviour of real-world problems and physical systems • Use logical reasoning to compare the utility of algorithms for the same problem • Collect and analyse data to meet the needs of known users		• Understand a range of ways to use technology safely, respectfully, responsibly and securely, including protecting their online identity and privacy • Recognise inappropriate content, contact and conduct and know how to report concerns